a11y

(Accessibility)



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- seit 2017 bei Coop in der internen Entwicklung
- 2019 – 2023 Bachelor an der FHNW
- 2025 – 2027 MSE (Master of Science in Engineering)

Stimmungsbild & Vorwissen



https://www.mentimeter.com/app/presentation/aly4y1xn5hvb8w5qznozzfjfjmd8pw1i/present?question=foc126kk186g

Arten von Einschränkungen

Permanent



One arm

Temporary



Broken arm

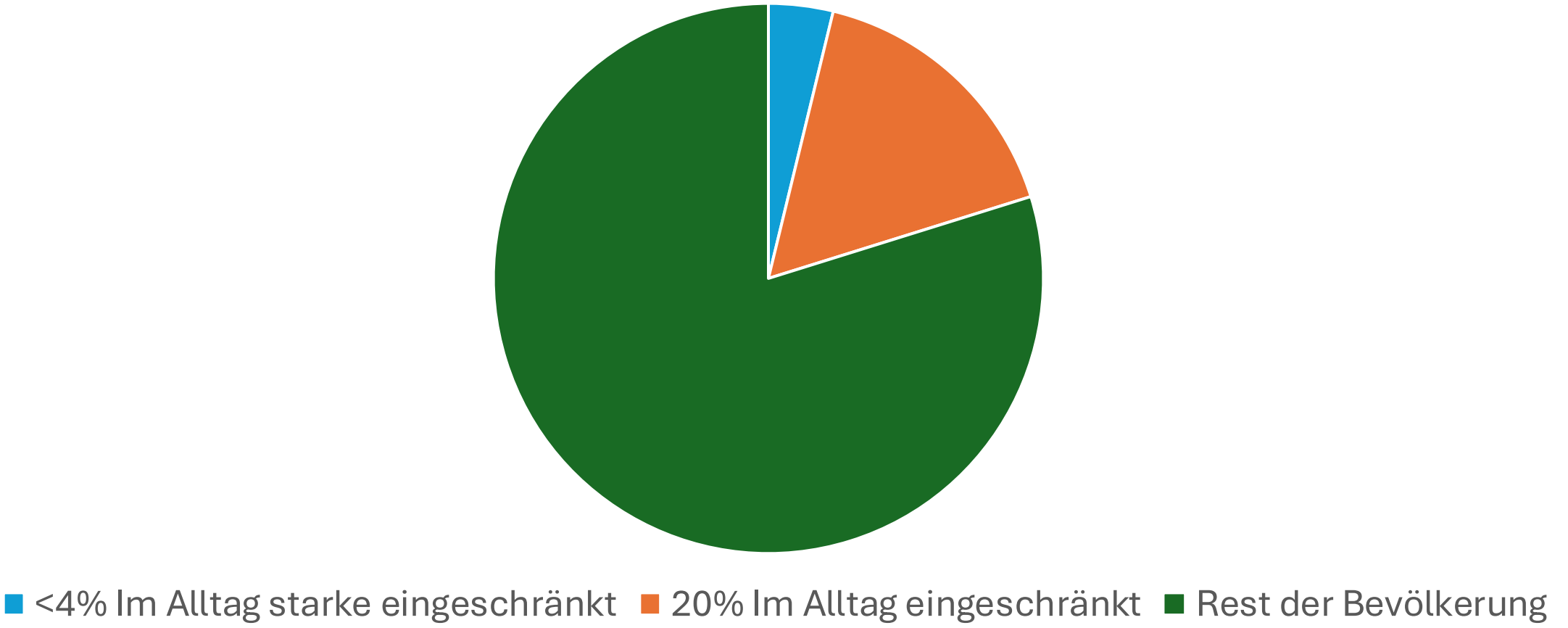
Situational



Holding a baby

https://uxdesign.cc/understanding-accessibility-makes-you-a-better-insert-your-name-here-1f7a15c50e7f

Schweizer Bevölkerung – 8’962’300 (permanente Beeinträchtigungen - Stand 2023)



https://www.bfs.admin.ch/bfs/de/home/statistiken/wirtschaftliche-soziale-situation-bevoelkerung/gleichstellung-menschen-behinderungen/behinderungen.html



Web-Applikationen für Accessibility (a11y) optimieren

Web-Applikationen optimieren, um 1ms schneller zu sein

https://en.meming.world/images/en/0/07/Drakeposting.jpg

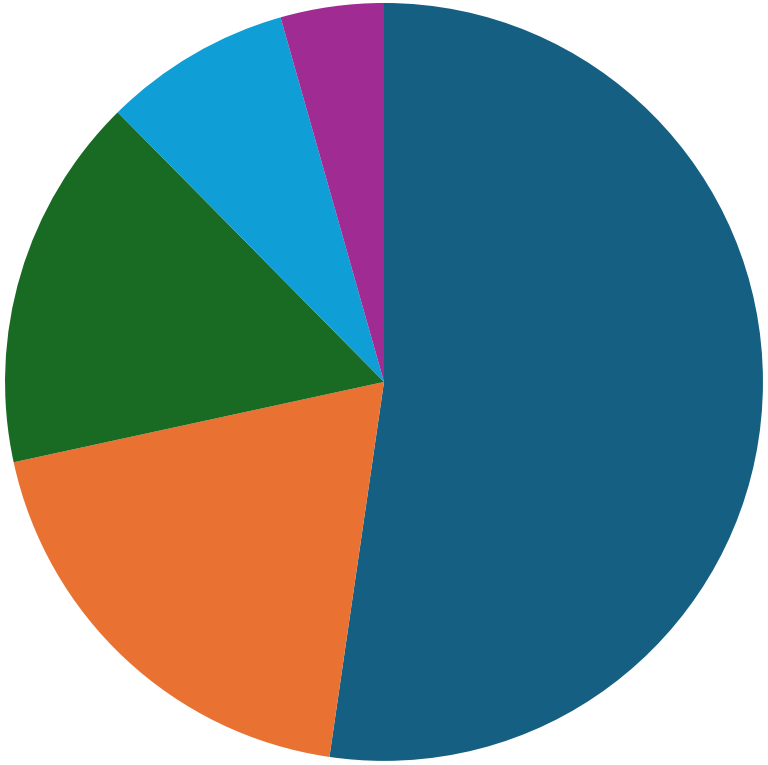
Jenseits von Maus und Bildschirm





https://i0.wp.com/css-tricks.com/wp-content/uploads/2022/08/jaws-nvda-voiceover.jpg
https://1lowvision.com/image/catalog/pacificvision/HIMS/BrailleEMotion/AngledRight-2-e1709748782159.png
https://resource.logitech.com/c_fill,q_auto,f_auto,dpr_1.0/d_transparent.gif/content/dam/logitech/en/products/keyboards/k250/gallery/delorean-gallery-images/deu/k-250-graphite-top-view-2-deu.png

Verwendete Browser in Kombination mit Screenreadern



Chrome 52.3% Microsoft Edge 19.3% Firefox 16% Safari 8% restliche Browser 4.4%

https://webaim.org/projects/screenreadersurvey10/

Semantik

Banane =



https://pngimg.com/uploads/banana/banana_PNG835.png

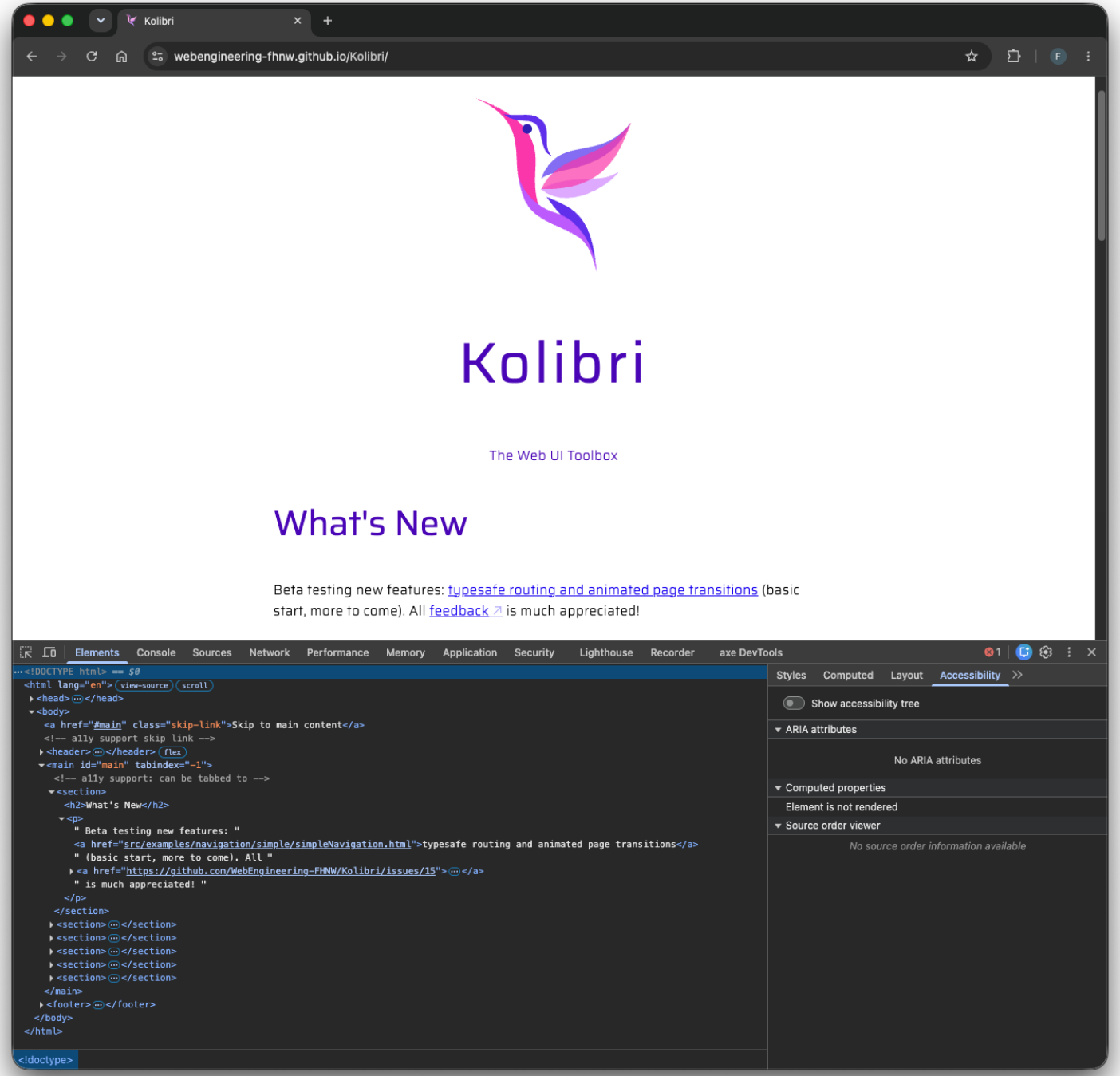
```
<button>Button Text</button>
```



vs

```
<div>Button Text</div>
```

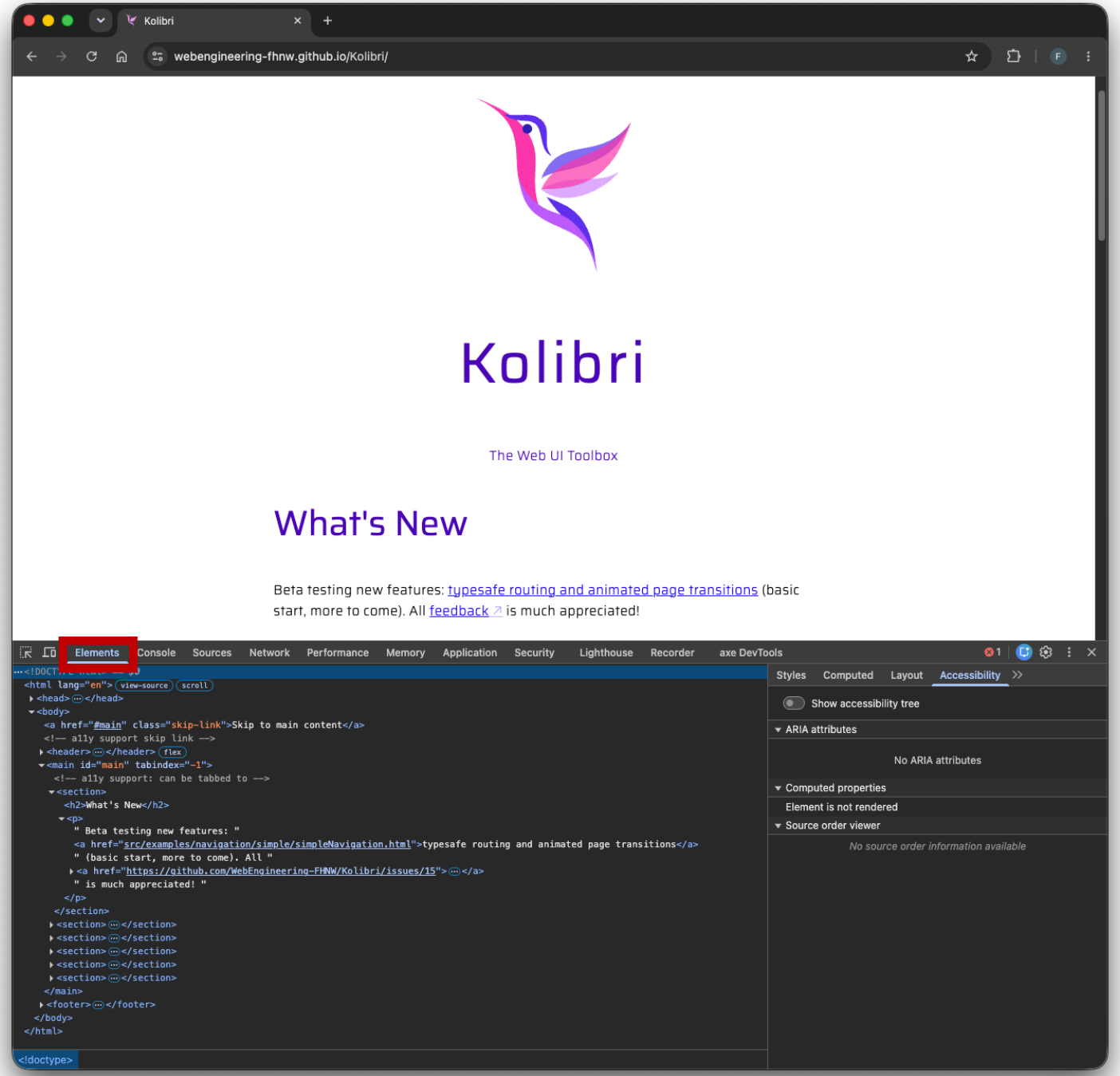
Button Text


Kolibri
webengineering-fhnw.github.io/Kolibri/
Kolibri
The Web UI Toolbox
What's New
Beta testing new features: typesafe routing and animated page transitions (basic start, more to come). All feedback is much appreciated!
Elements Console Sources Network Performance Memory Application Security Lighthouse Recorder axe DevTools
<!DOCTYPE html> == $0
<html lang="en"> view-source scroll
<head> </head>
<body>
Skip to main content
<!-- a11y support skip link -->
<header> </header> flex
<main id="main" tabindex="-1">
<!-- a11y support: can be tabbed to -->
<section>
<h2>What's New</h2>

" Beta testing new features: "
typesafe routing and animated page transitions
" (basic start, more to come). All "

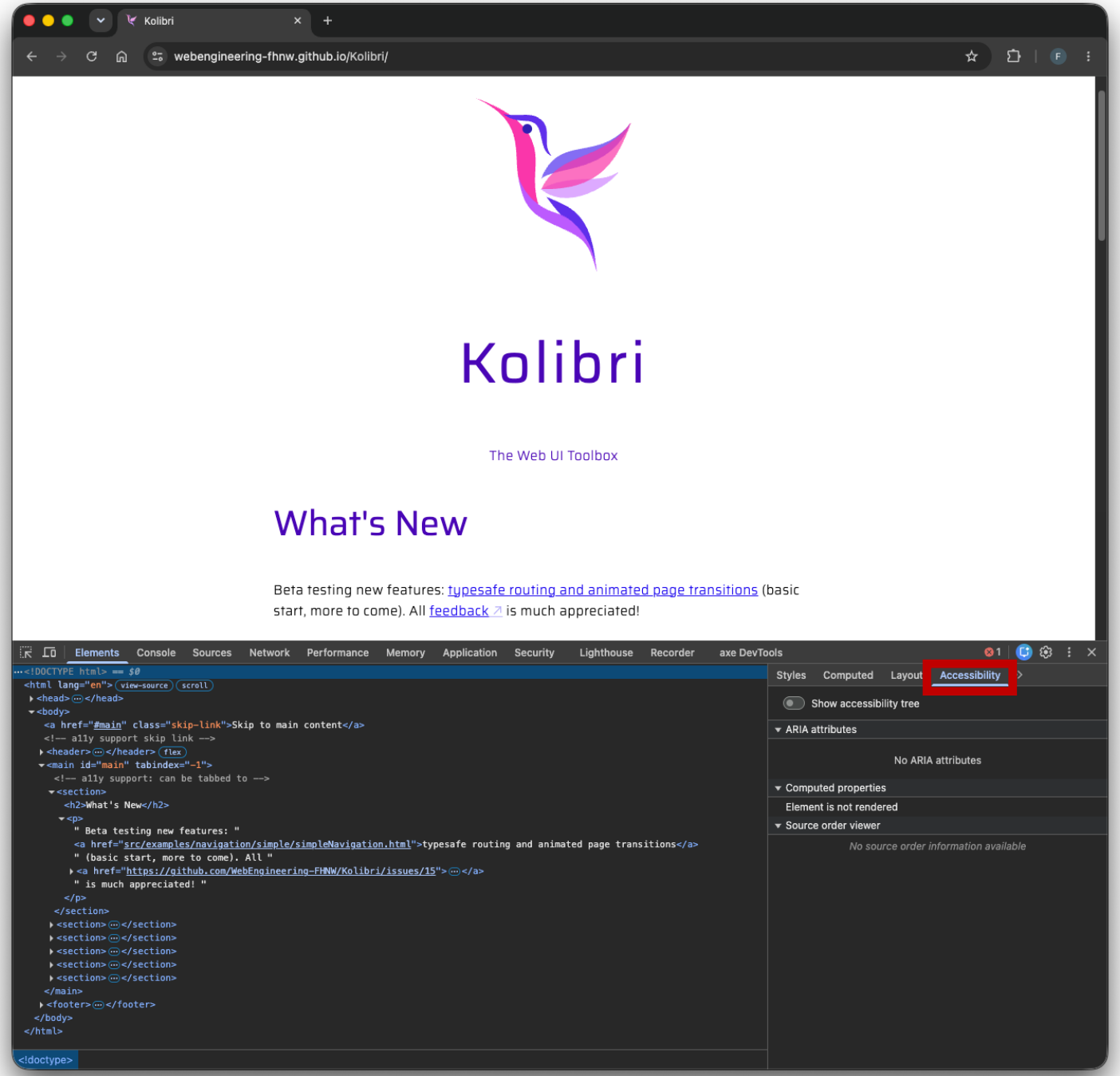
" is much appreciated! "

</section>
<section> </section>
<section> </section>
<section> </section>
<section> </section>
<section> </section>
</main>
<footer> </footer>
</body>
</html>
<!doctype>
Styles Computed Layout Accessibility
Show accessibility tree
ARIA attributes
No ARIA attributes
Computed properties
Element is not rendered
Source order viewer
No source order information available


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<a href="https://github.com/WebEngineering-FHNW/Kolibri/issues/15">
" is much appreciated! "

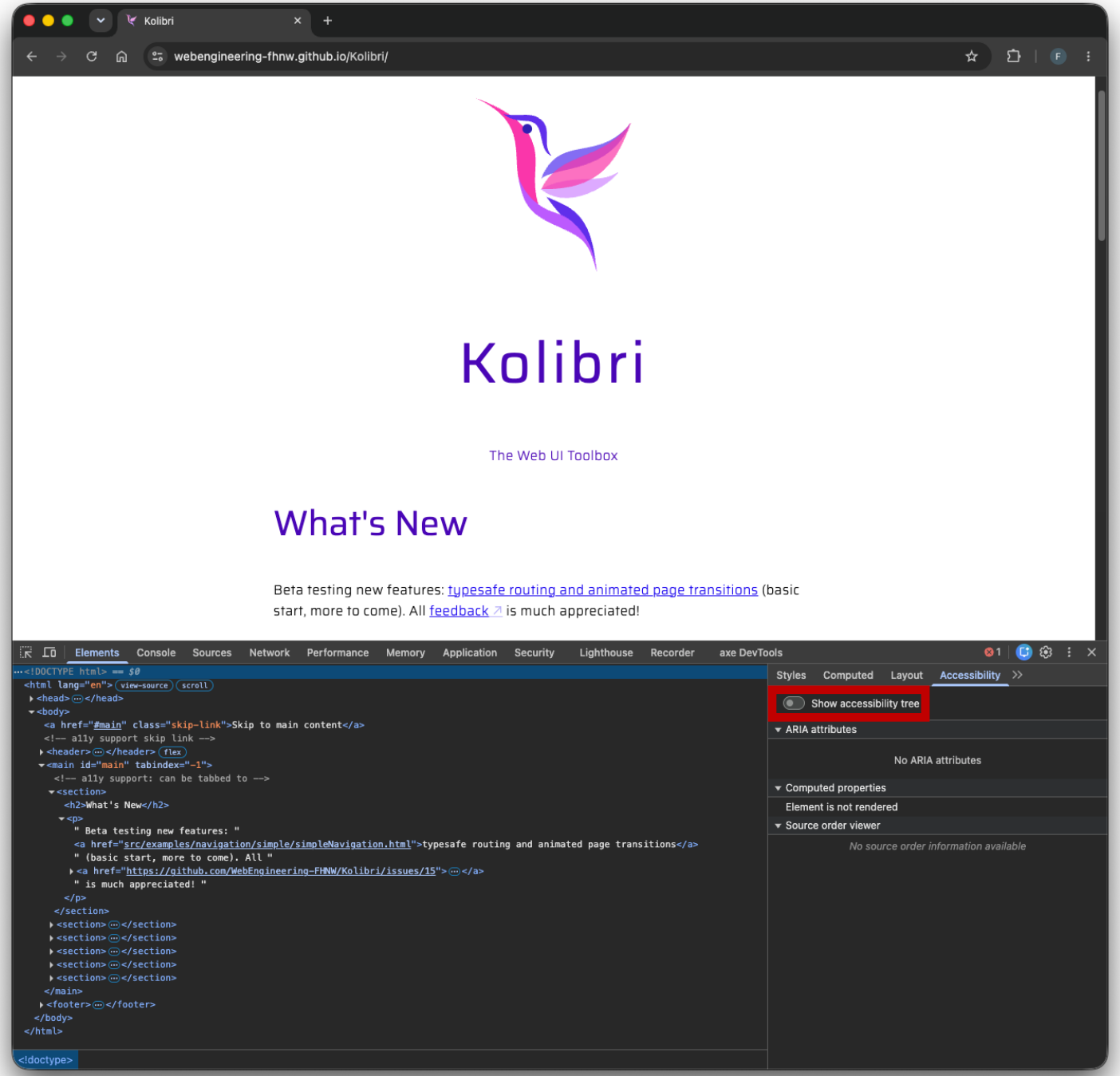
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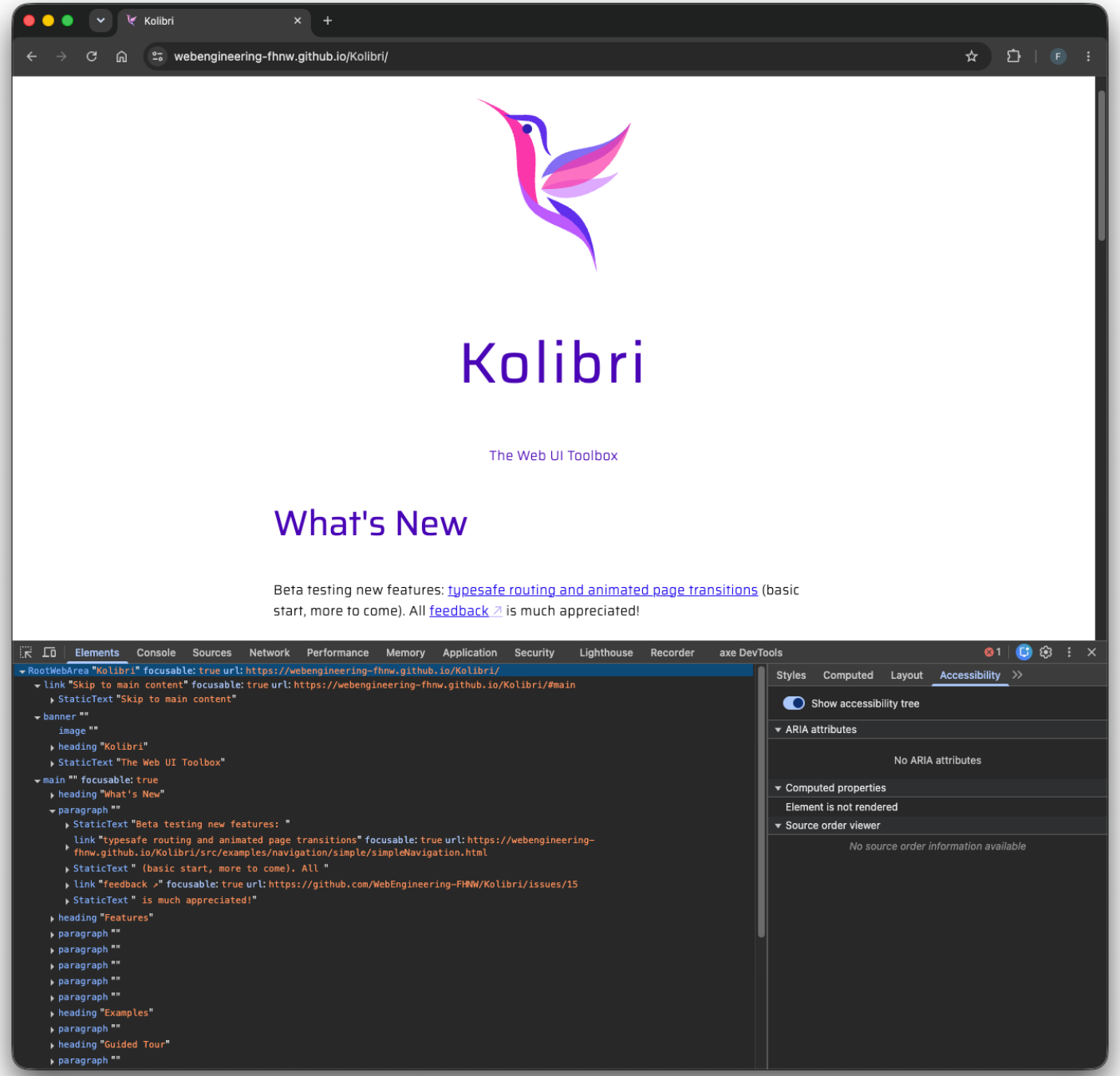
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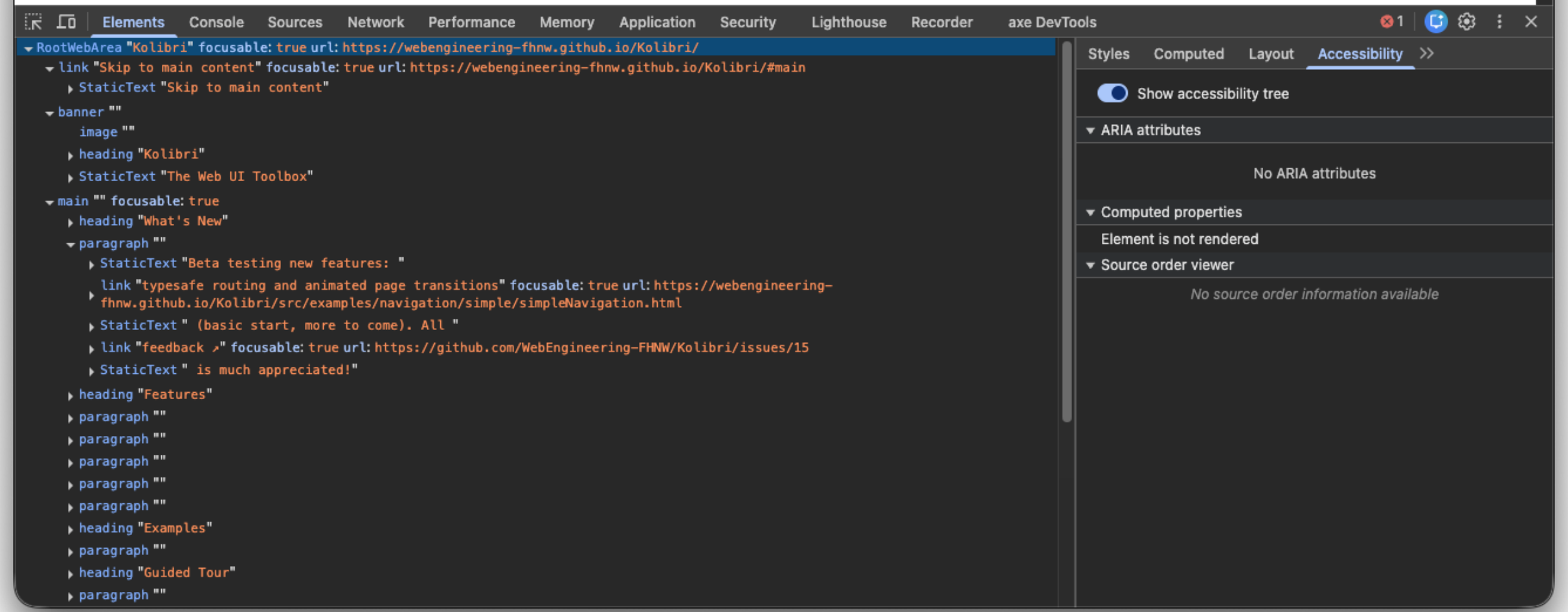
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" (basic start, more to come). All "

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RootWebArea "Kolibri" focusable: true url: https://webengineering-fhnw.github.io/Kolibri/
link "Skip to main content" focusable: true url: https://webengineering-fhnw.github.io/Kolibri/#main
StaticText "Skip to main content"
banner ""
image ""
heading "Kolibri"
StaticText "The Web UI Toolbox"
main "" focusable: true
heading "What's New"
paragraph ""
StaticText "Beta testing new features: "
link "typesafe routing and animated page transitions" focusable: true url: https://webengineering-fhnw.github.io/Kolibri/src/examples/navigation/simple/simpleNavigation.html
StaticText " (basic start, more to come). All "
link "feedback ↗" focusable: true url: https://github.com/WebEngineering-FHNW/Kolibri/issues/15
StaticText " is much appreciated!"
heading "Features"
paragraph ""
paragraph ""
paragraph ""
paragraph ""
paragraph ""
heading "Examples"
paragraph ""
heading "Guided Tour"
paragraph ""
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```
<button>Button Text</button>
```

```
button "Button Text" focusable: true
  StaticText "Button Text"
    InlineTextBox "Button Text"
```

VS

```
<div>Button Text</div>
```

```
StaticText "Button Text"
  InlineTextBox "Button Text"
```

▶ Berry fun fact

▼ Berry fun fact

Botanically speaking, bananas, watermelons, and pumpkins are all true berries, but strawberries and raspberries aren't.

```
<details name="facts">
  <summary>
    Berry fun fact
  </summary>
  Botanically speaking, bananas, watermelons, and
  pumpkins are all true berries, but strawberries
  and raspberries aren't.
</details>
```

▶ Berry fun fact

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```

▶ Berry fun fact

▼ Berry fun fact

Botanically speaking, bananas, watermelons, and pumpkins are all true berries, but strawberries and raspberries aren't.

Uploading Document



```
<label for="progress-bar">
  Uploading Document
</label>
<progress id="progress-bar" value="70" max="100">
  70%
</progress>
```

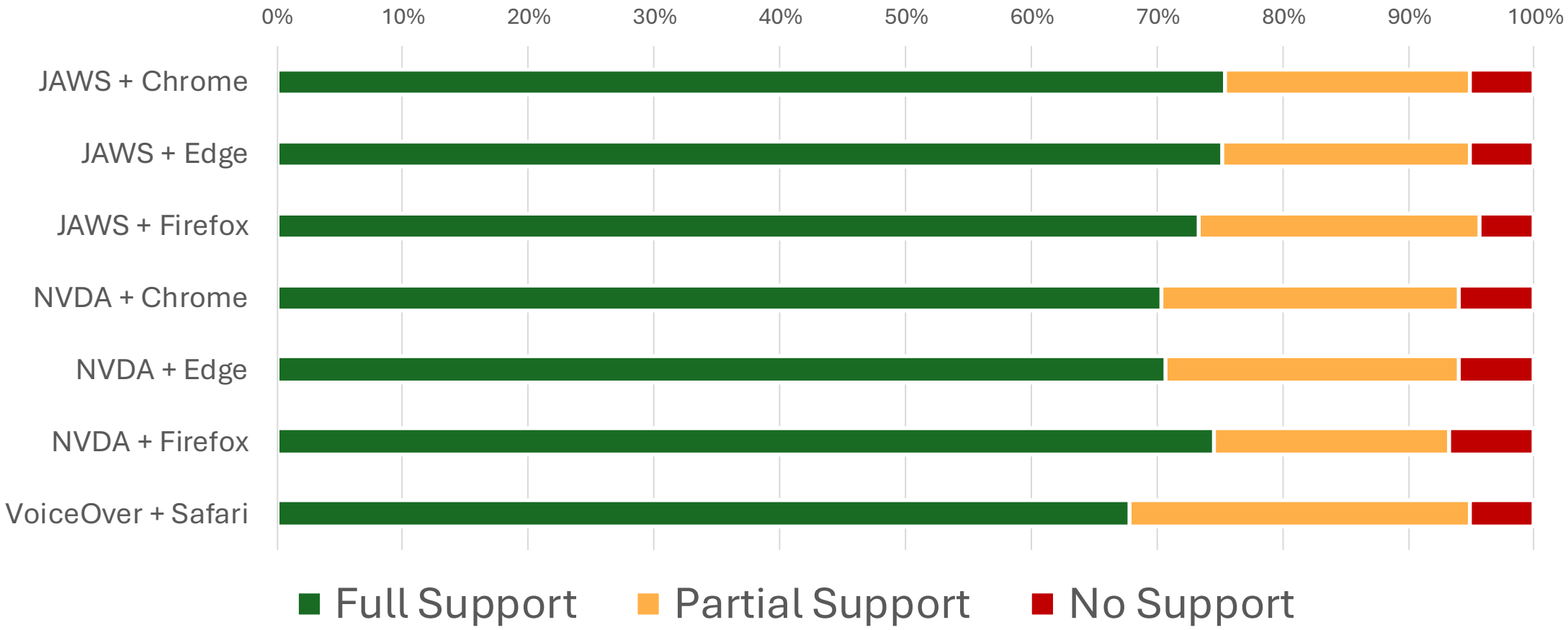


ARIA (Accessible Rich Internet Applications)

First Rule of ARIA Use

"*If you can use a native HTML element or attribute with the semantics and behavior you require* ***already built in****, instead of re-purposing an element and adding an ARIA role, state or property to make it accessible****, then do so***"

ARIA Support



https://a11ysupport.io/

aria-hidden

```
<button type="submit">
  <span class="label">Send Message</span>
  <span class="icon" aria-hidden="true">&#x27A4;</span>
</button>
```



aria-hidden

```
<button type="submit">
  <span class="label">Send Message</span>
  <span class="icon" aria-hidden="true">&#x27A4;</span>
</button>
```



“Send Message Black Rightwards Arrowhead”

aria-label

```
<button type="submit" aria-label="Send message">
  <span class="icon" aria-hidden="true">&#xFE0F;</span>
</button>
```



aria-label

```
<button type="submit" aria-label="Send message">
  <span class="icon" aria-hidden="true">&#xFE0F;</span>
</button>
```



“Send Message”

Bessere Lösung ohne ARIA

```
<button type="submit" class="btn-send"></button>
```

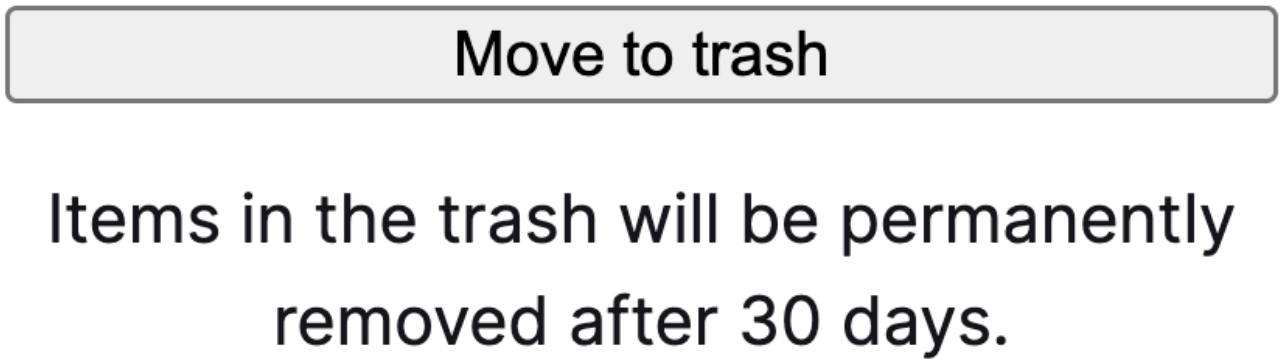
```
.btn-send::before {
  content: "\2709\fe0f" / "Send message";
}
```



“Send Message”

aria-describedby

```
<button aria-describedby="trash-desc">Move to trash</button>
<p id="trash-desc">
  Items in the trash will be permanently
  removed after 30 days.
</p>
```



Semantik in Eigenverantwortung

```
<div
  id="my-button"
  class="button"
  role="button"
  tabindex="0">
  Button Text
</div>
```

Button Text

```css
.button {
  display:          inline-block;
  padding:          2px 7px;
  font:             -webkit-small-control;
  background-color: #EFEFEF;
  border:           1px solid #767676;
  border-radius:    2px;
  cursor:           default;
}

.button:hover {
  background-color: #E5E5E5;
  border-color:     #666666;
}
```

Original vs Fake





```
const button = document.getElementById('my-button');

function handleAction() {
  //Business logic
}

button.addEventListener('click', handleAction);

button.addEventListener('keydown', event => {
  if (event.key === 'Enter' || event.key === ' ') {
    event.preventDefault();
    handleAction();
  }
});
```

WCAG (Web Content Accessibility Guideline)



https://www.w3.org/WAI/WCAG22/wcag2.2AAA-blue.svg

1

Level A (Grundanforderungen)

- Bedeutet eine Seite ist nicht komplett unzugänglich.

2

Level AA (Industrie Standard)

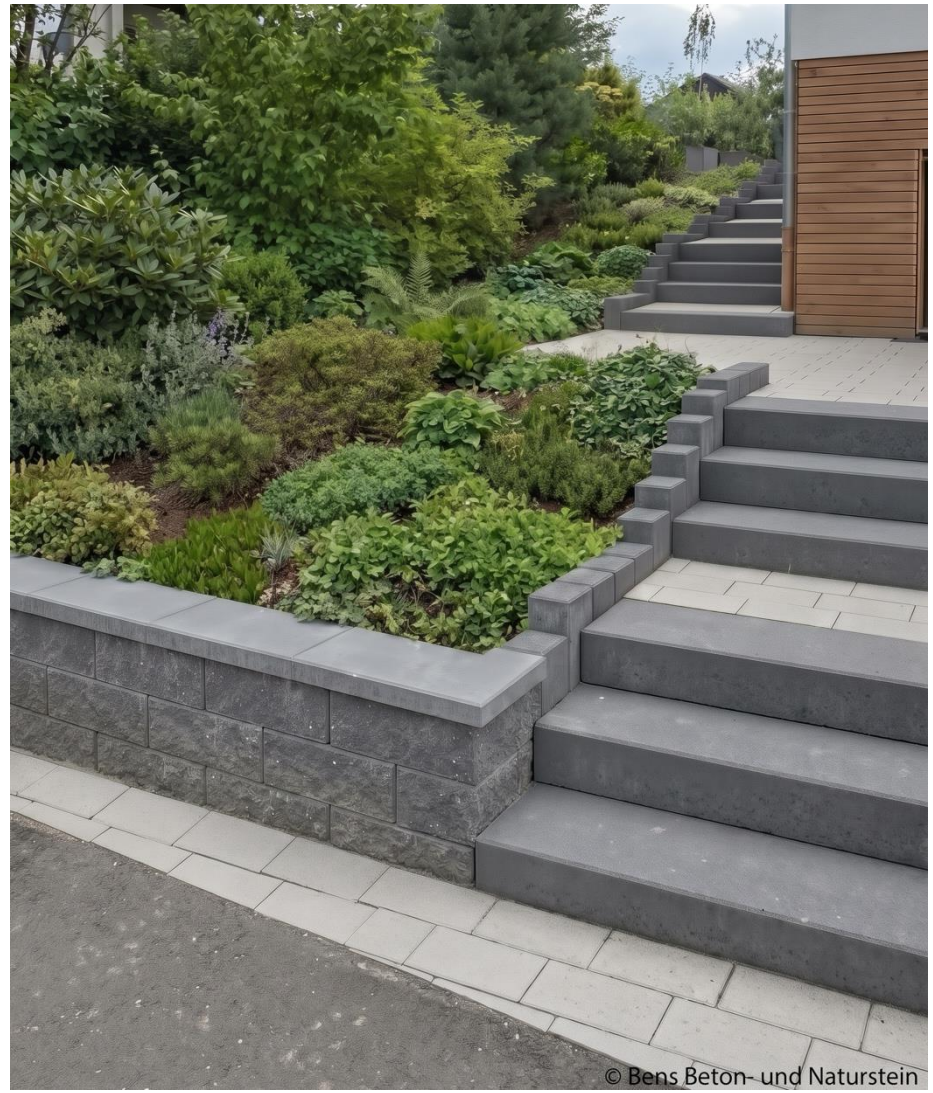
- Standard, auf den sich auch der European Accessibility Act bezieht. (EN 301 549)

3

Level AAA (Gold Standard)

- Kontextbedingt und nicht auf jeden Content anwendbar.

https://www.barrierefreiheit-dienstekonsolidierung.bund.de/Webs/PB/DE/gesetze-und-richtlinien/en301549/en301549-node.html


© Bens Beton- und Naturstein

Level A

Sinnbildliches Alltagsbeispiel



Analoges Web-Beispiel



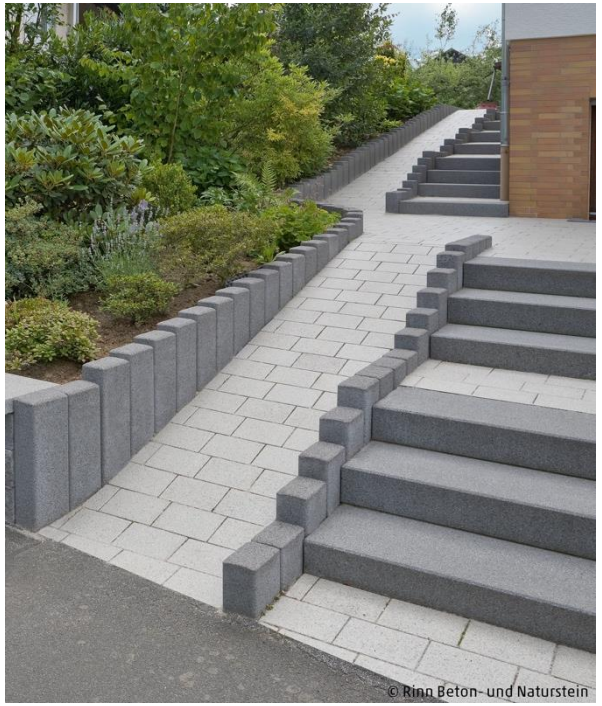

```
<img
  src="/path/to/lemon.webp"
  alt="Ganze Bio-Zitrone
       mit grünem Blatt" />
```

Success Criterion 1.1.1 Non-text Content

https://i.pinimg.com/originals/f6/a7/83/f6a78356f703f7ecb63e61eec668998e.jpg
https://media01.stockfood.com/largepreviews/NDkxMjUwOA==/00158468-Eine-Zitrone-mit-Blatt.jpg

Level A

Sinnbildliches Alltagsbeispiel



Analoges Web-Beispiel




```
<img
  src="/path/to/lemon.webp"
  alt="Ganze Bio-Zitrone
       mit grünem Blatt" />
```

Success Criterion 1.1.1 Non-text Content

https://i.pinimg.com/originals/f6/a7/83/f6a78356f703f7ecb63e61eec668998e.jpg
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Level AA

Sinnbildliches Alltagsbeispiel



Analoges Web-Beispiel

Text mit 4.5:1 Kontrast Ratio

```
p {
  background-color: #FFFFFF;
  color:            #BF25E9;
}
```

Success Criterion 1.4.3 Contrast (Minimum)

$$L = 0.2126 \cdot R + 0.7152 \cdot G + 0.0722 \cdot B$$

$$\text{Where } C = \begin{cases} \frac{C_{\text{sRGB}}}{12.92} & \text{if } C_{\text{sRGB}} \leq 0.03928 \\ \left(\frac{C_{\text{sRGB}}+0.055}{1.055}\right)^{2.4} & \text{otherwise} \end{cases}$$

Example: Dodger Blue `rgb(30, 144, 255)`

1. Normalize (/255)

$R_{\text{sRGB}} = \frac{30}{255} = 0.1176$
$G_{\text{sRGB}} = \frac{144}{255} = 0.5647$
$B_{\text{sRGB}} = \frac{255}{255} = 1.0000$

2. Linearize

$R = 0.0123$
$G = 0.2747$
$B = 1.0000$

3. Calculate Luminance (L)

$$L = (0.2126 \cdot 0.0123) + (0.7152 \cdot 0.2747) + (0.0722 \cdot 1.0000) = \mathbf{0.2712}$$

Calculate Contrast Ratio

$$\text{Ratio} = \frac{\text{L1} + 0.05}{\text{L2} + 0.05}$$

$$L = 0.2126 \cdot R + 0.7152 \cdot G + 0.0722 \cdot B$$

$$\text{Where } C = \begin{cases} \frac{C_{\text{sRGB}}}{12.92} & \text{if } C_{\text{sRGB}} \leq \sout{0.03928} \text{ 0.04045} \\ \left(\frac{C_{\text{sRGB}}+0.055}{1.055}\right)^{2.4} & \text{otherwise} \end{cases}$$

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1. Normalize (/255)

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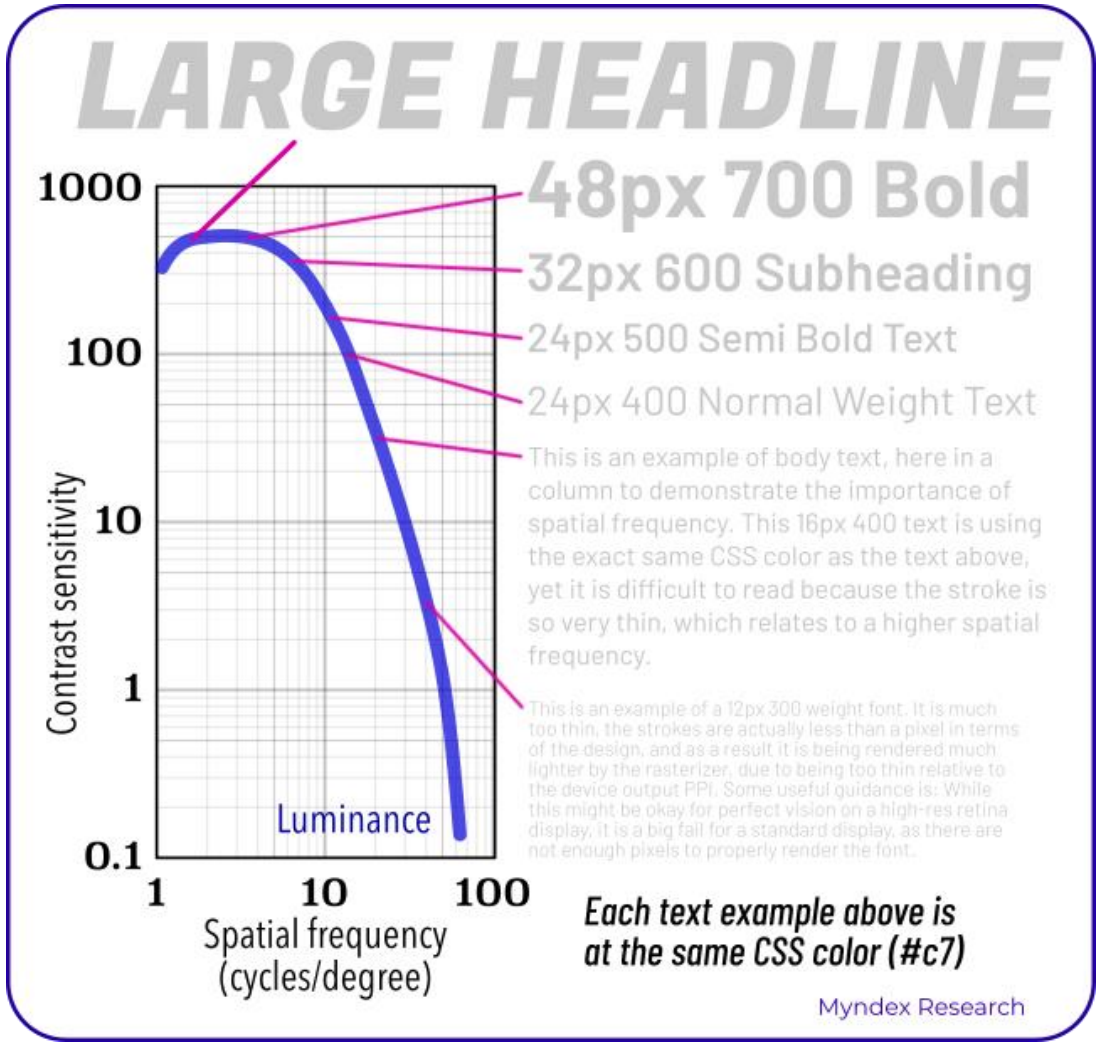
3. Calculate Luminance (L)

$$L = (0.2126 \cdot 0.0123) + (0.7152 \cdot 0.2747) + (0.0722 \cdot 1.0000) = \mathbf{0.2712}$$

Calculate Contrast Ratio

$$\text{Ratio} = \frac{\text{L1} + 0.05}{\text{L2} + 0.05}$$

APCA (Accessible Perceptual Contrast Algorithm)



https://git.apcacontrast.com/documentation/WhyAPCA.html

Level AA

Sinnbildliches Alltagsbeispiel



Analoges Web-Beispiel

Text mit 4.5:1 Kontrast Ratio

```
p {
  background-color: #FFFFFF;
  color:            #BF25E9;
}
```

Success Criterion 1.4.3 Contrast (Minimum)

Level AAA

Sinnbildliches Alltagsbeispiel



Analoges Web-Beispiel

Breadcrumbs > showing > **your Location**

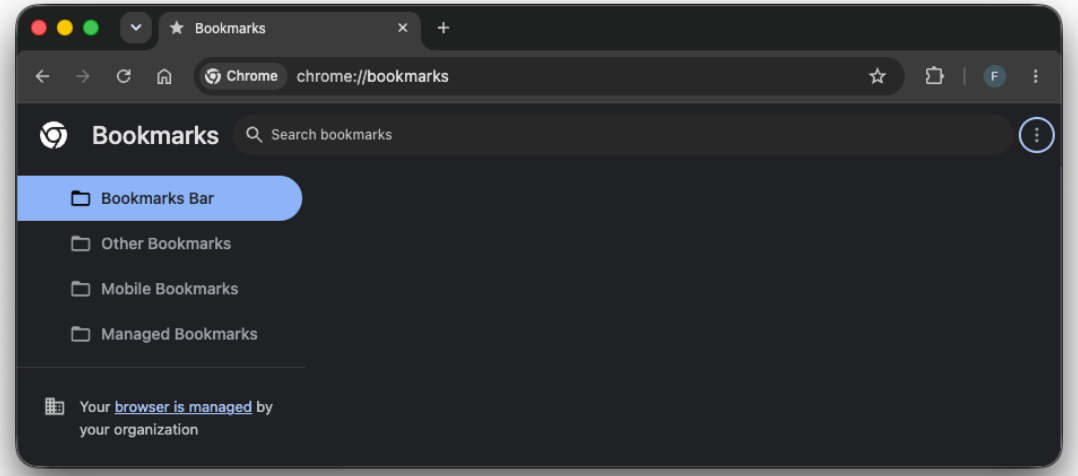
Success Criterion 2.4.8 Location

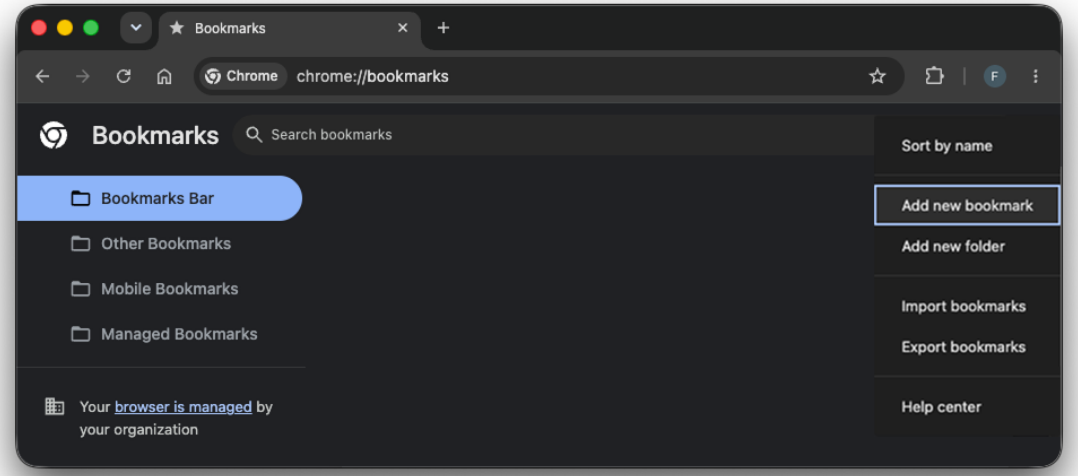
Hands on

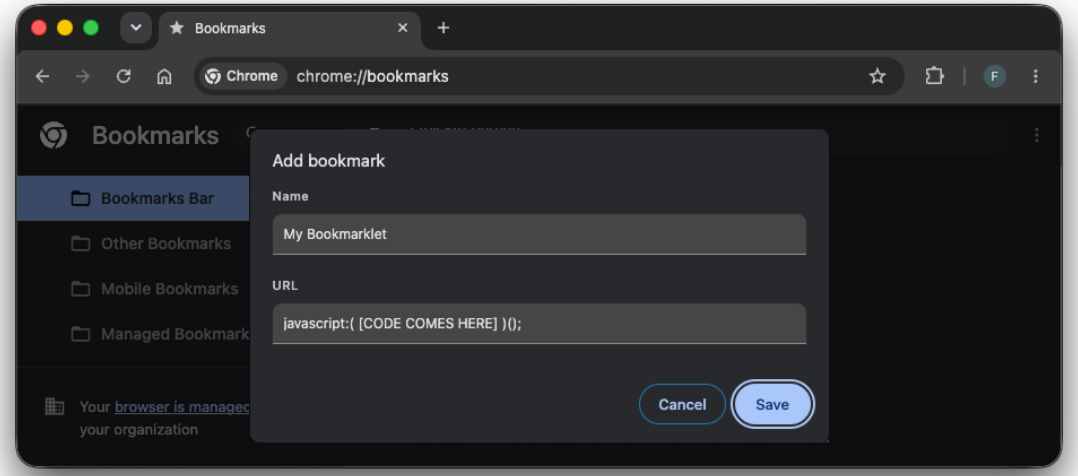
Visual Impairment Übung

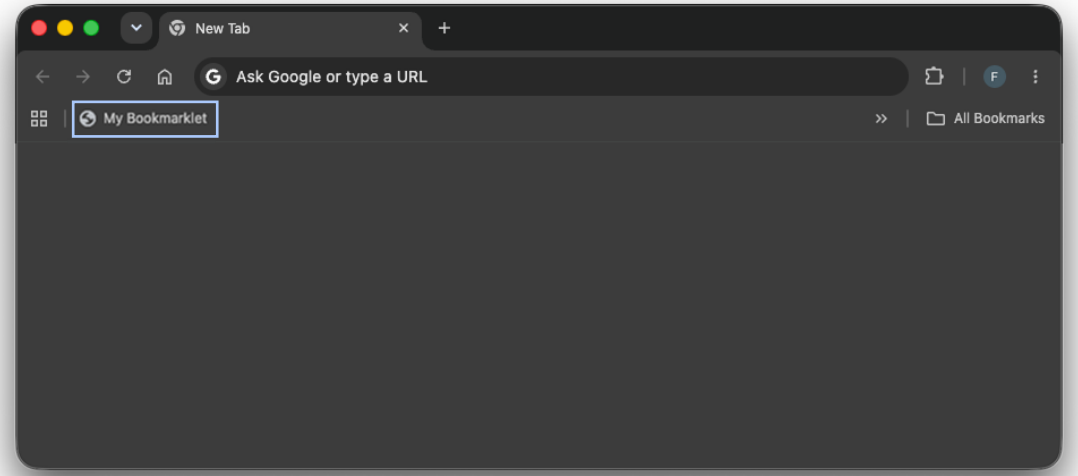
https://github.com/Flaverus/a11y

Bookmarklet - https://github.com/Flaverus/a11y

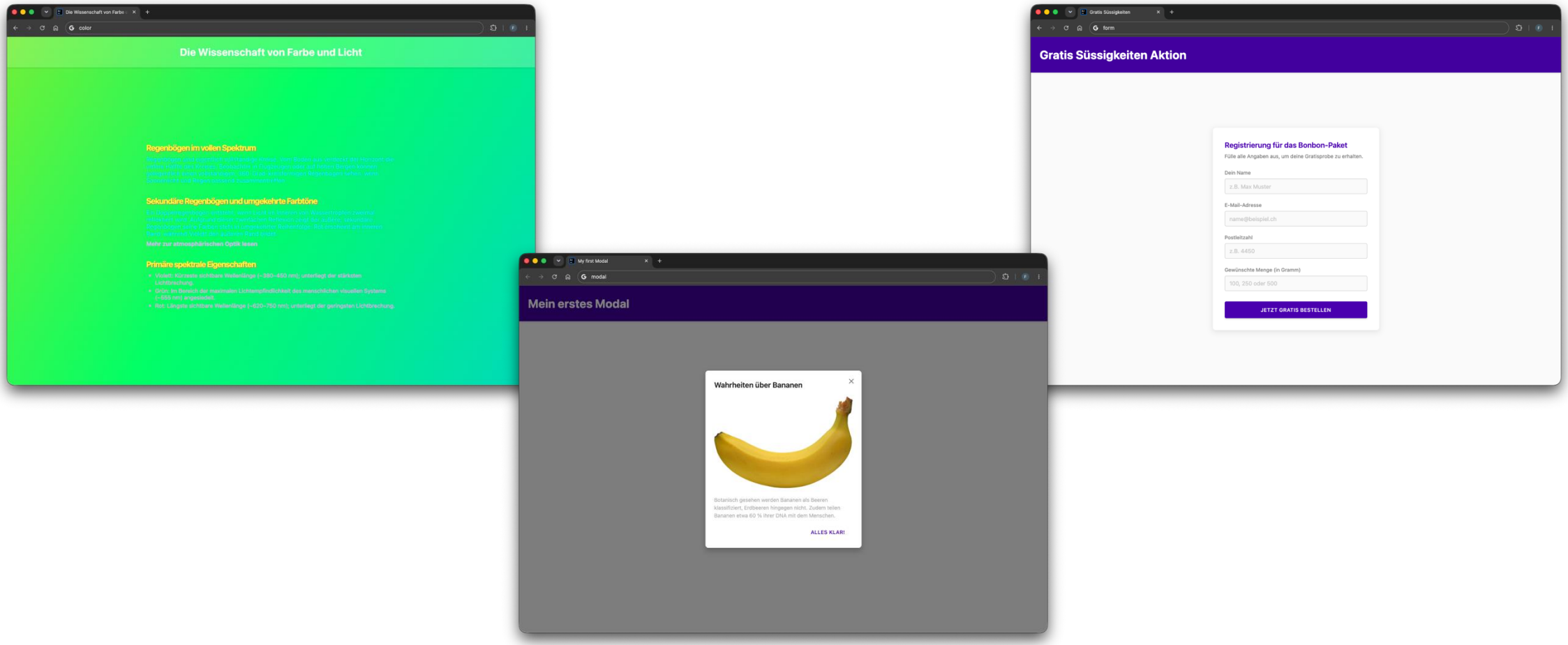




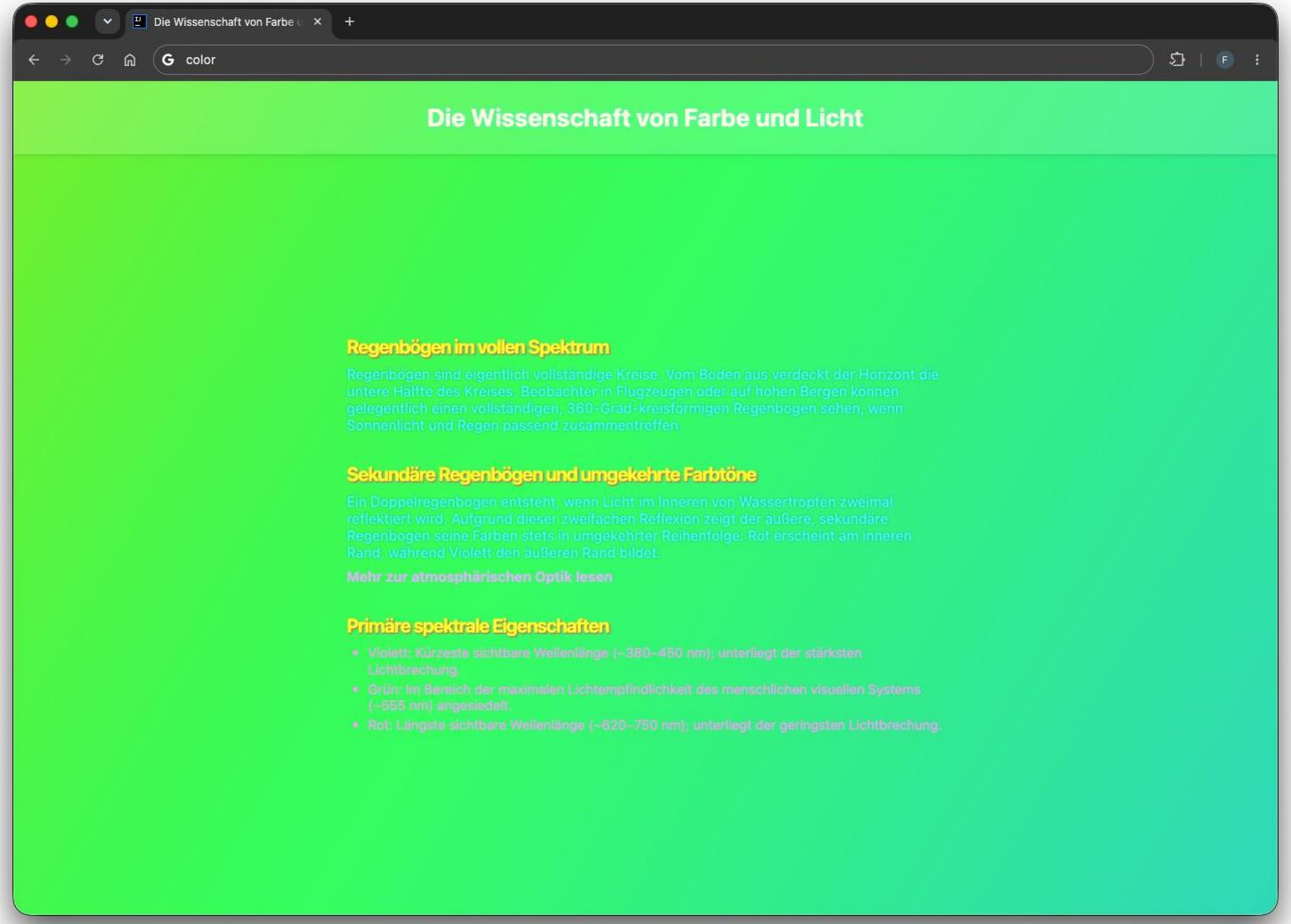




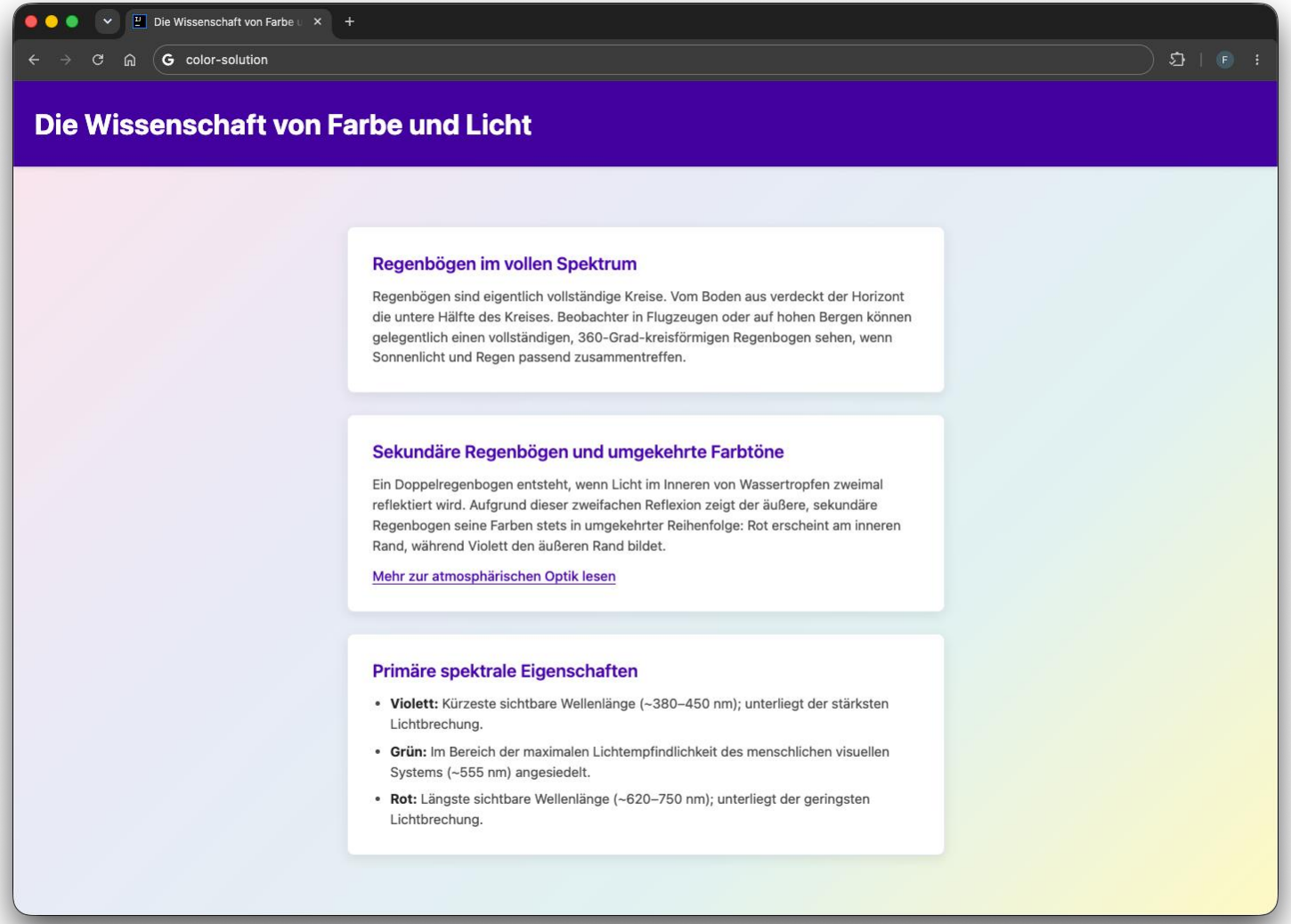
Seiten - https://github.com/Flaverus/a11y



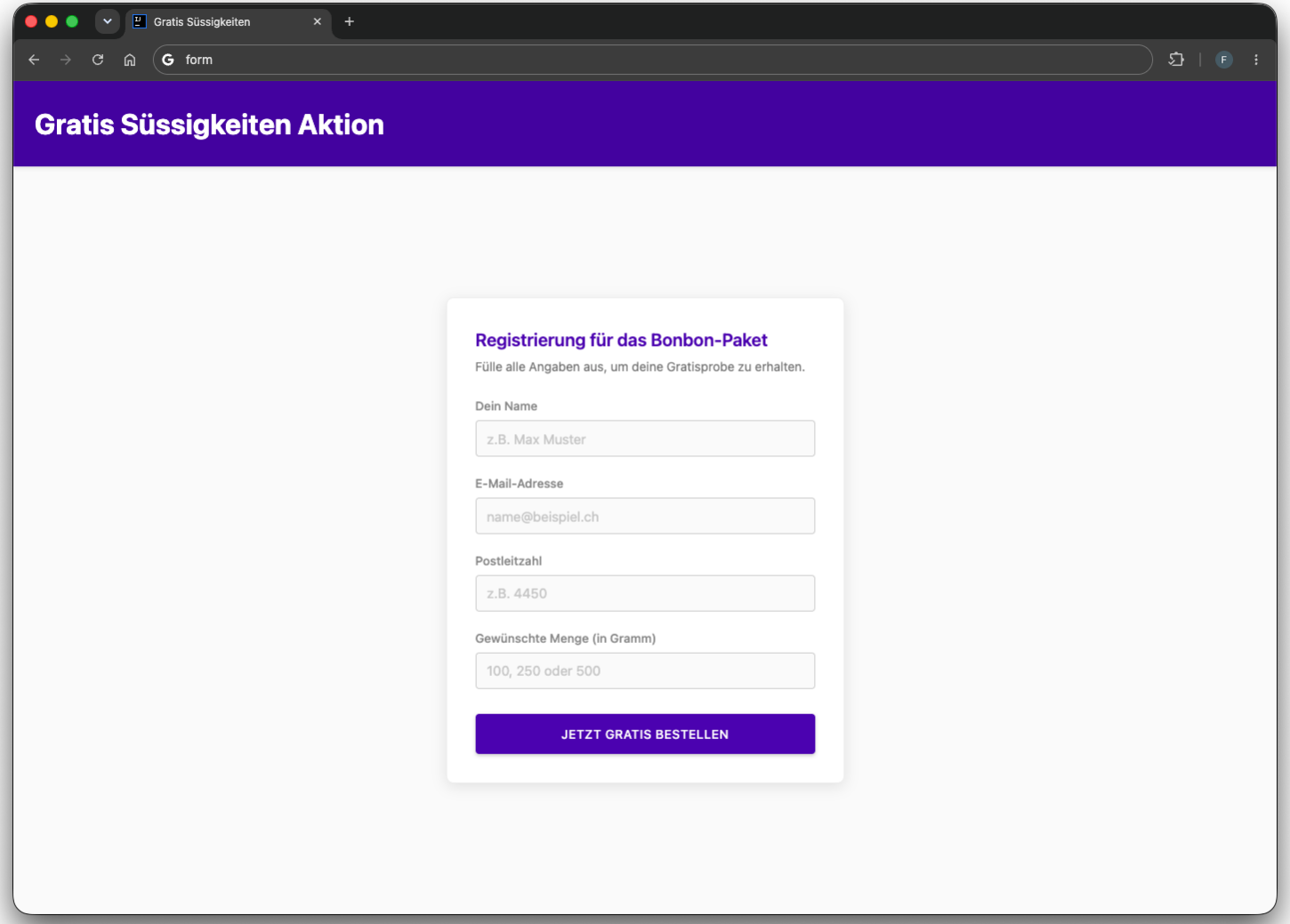
Colors


Die Wissenschaft von Farbe
color
Die Wissenschaft von Farbe und Licht
Regenbögen im vollen Spektrum
Regenbögen sind eigentlich vollständige Kreise. Vom Boden aus verdeckt der Horizont die untere Hälfte des Kreises. Beobachter in Flugzeugen oder auf hohen Bergen können gelegentlich einen vollständigen, 360-Grad-kreisförmigen Regenbogen sehen, wenn Sonnenlicht und Regen passend zusammentreffen.
Sekundäre Regenbögen und umgekehrte Farbtöne
Ein Doppelregenbogen entsteht, wenn Licht im Inneren von Wassertropfen zweimal reflektiert wird. Aufgrund dieser zweifachen Reflexion zeigt der äußere, sekundäre Regenbogen seine Farben stets in umgekehrter Reihenfolge: Rot erscheint am inneren Rand, während Violett den äußeren Rand bildet.
Mehr zur atmosphärischen Optik lesen
Primäre spektrale Eigenschaften
Violett: Kürzeste sichtbare Wellenlänge (~380–450 nm); unterliegt der stärksten Lichtbrechung.
Grün: Im Bereich der maximalen Lichtempfindlichkeit des menschlichen visuellen Systems (~555 nm) angesiedelt.
Rot: Längste sichtbare Wellenlänge (~620–750 nm); unterliegt der geringsten Lichtbrechung.

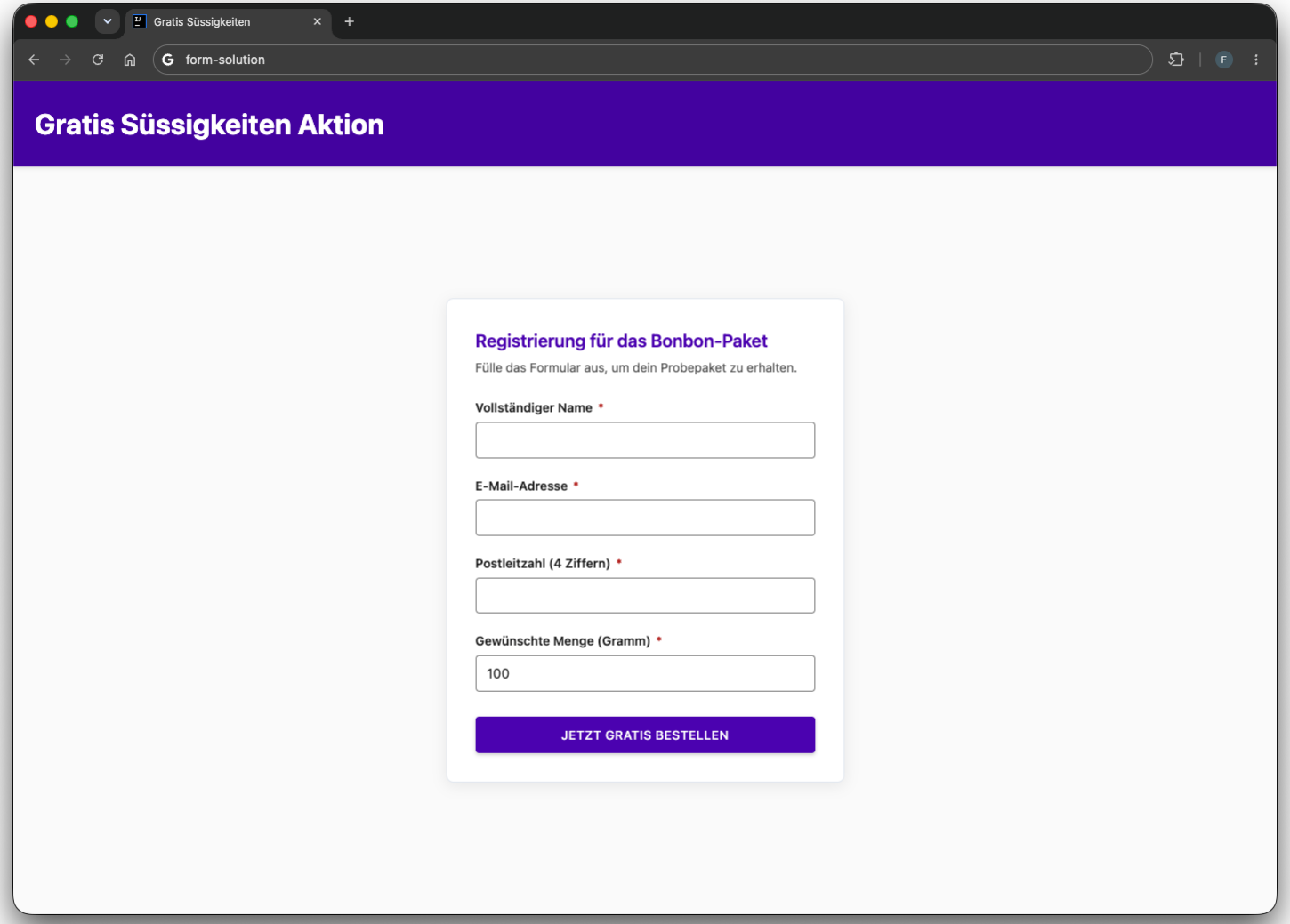
Colors


Die Wissenschaft von Farbe
color-solution
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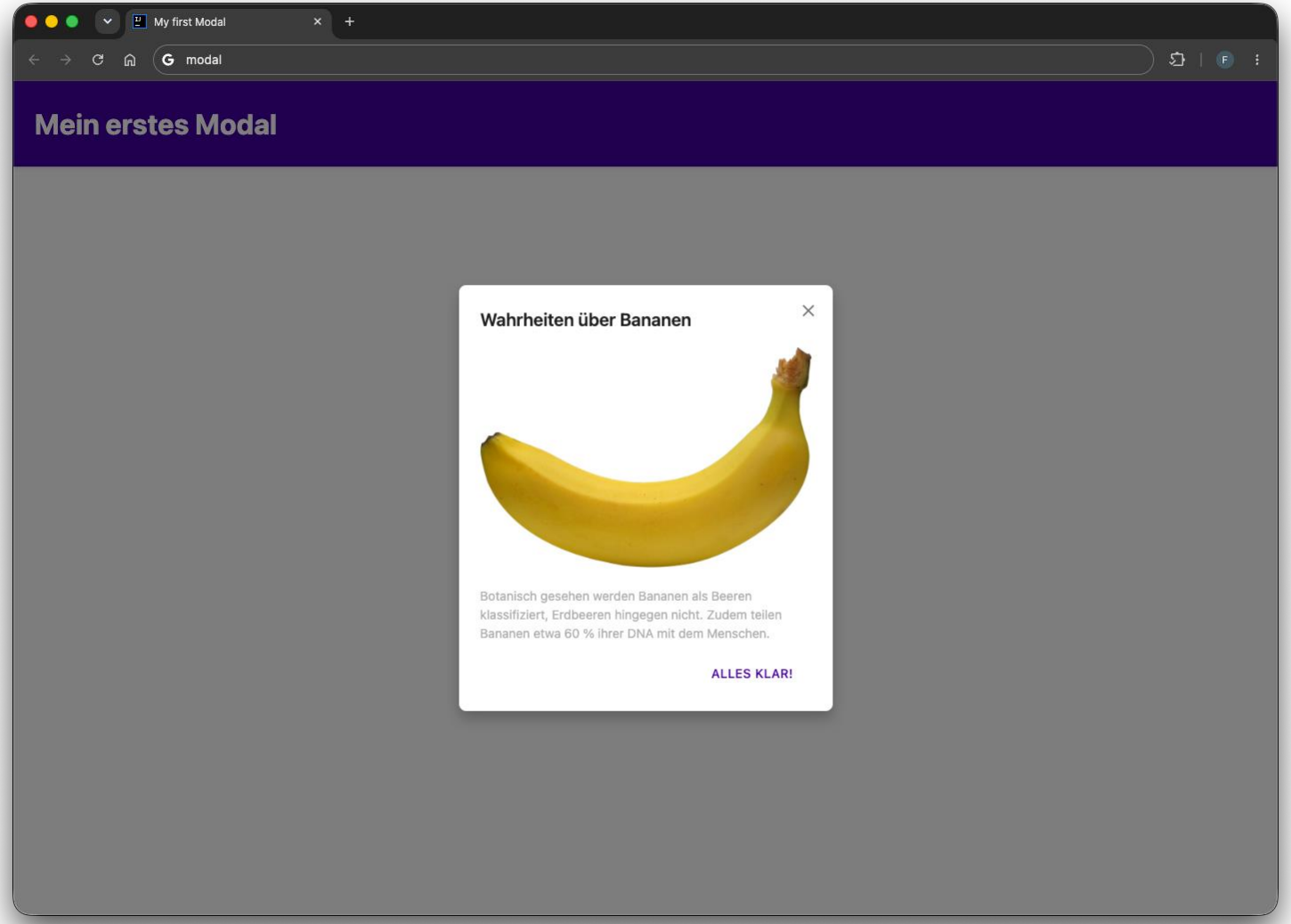
Form


Gratis Süssigkeiten
form
Gratis Süssigkeiten Aktion
Registrierung für das Bonbon-Paket
Fülle alle Angaben aus, um deine Gratisprobe zu erhalten.
Dein Name
z.B. Max Muster
E-Mail-Adresse
name@beispiel.ch
Postleitzahl
z.B. 4450
Gewünschte Menge (in Gramm)
100, 250 oder 500
JETZT GRATIS BESTELLEN

Form


Gratis Süssigkeiten
form-solution
Gratis Süssigkeiten Aktion
Registrierung für das Bonbon-Paket
Fülle das Formular aus, um dein Probepaket zu erhalten.
Vollständiger Name *
E-Mail-Adresse *
Postleitzahl (4 Ziffern) *
Gewünschte Menge (Gramm) *
100
JETZT GRATIS BESTELLEN

Modal


My first Modal
modal
Mein erstes Modal
Wahrheiten über Bananen
Botanisch gesehen werden Bananen als Beeren klassifiziert, Erdbeeren hingegen nicht. Zudem teilen Bananen etwa 60 % ihrer DNA mit dem Menschen.
ALLES KLAR!

Modal

```
<button
  type="button"
  class="btn btn-primary"
  commandfor="fruit-dialog"
  command="show-modal">
  Discover Fruit Fact
</button>

<dialog id="fruit-dialog" closedby="any">
  <!-- Modal Contents -->
</dialog>
```



https://create.kahoot.it/creator/ce963005-a417-4cb4-beb6-484acac60fd1

Neue Erkentnisse
Anregungen
Themenwünsche

Danke für eure Aufmerksamkeit!